MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Underdrainage For Tailings

MD-101-SMEM-EG-CV-SPC-0010

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Table of Contents

1. Introduction 3

1.1 Scope 3
1.2 Purpose 3

2. Key Terms and Definitions 3

3. Referenced Documents 4

3.1 Statutory Regulations 4
3.2 Industry Codes and Standards 4
3.2.1 International Codes and Standards 4
3.2.2 Alternative Codes 5
3.3 Ma'aden Standards 5

4. Materials 5

4.1 General 5
4.2 Corrugated Perforated Drainage Pipe Material 5
4.2.1 Pipe Strength 5
4.2.2 Pipe Perforations 5
4.2.3 Filter Fabric 6
4.3 Packaging and Handling 6
4.3.1 Drain-Pipe Coils 6
4.3.2 Rigid Pipe (with Geotextile) 6
4.4 Transportation, Loading and Unloading 7
4.5 Quality Assurance 7
4.5.1 General 7
4.5.2 Quality Assurance of Materials 7
4.6 Tolerances 8

5. Installation 8

5.1 General 8
5.2 Backfill Material 8
5.3 Flushing 8
5.4 Underdrainage 9
5.5 Jointing 9
5.6 Backfilling 9
5.7 Tolerances 9

6. Submittals 9

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for supply and installation of underdrainage for a tailing storage area for all Ma'aden projects.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

The purpose of this specification is to:

- Identify applicable Standards and guidelines to be used for the Work, and
- Set the minimum engineering standard for supply and installation of underdrainage for all Ma'aden projects.

The designer is required to specify any requirements unique to the tailings being impounded. The designer shall be responsible for checking the chemical compatibilities and physical properties of the materials. Typical considerations include:

1. Tailings for alumina production – Caustic renders polyester un-suitable.
2. Tailings for phosphate production can be high in gypsum. Geotextile filters are not appropriate if there is a potential of clogging.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Lining system	Material or combination of materials designed and constructed as a low permeability barrier for liquids or gases.
PE	Polyethylene
PVC	Poly Vinyl Chloride
Site	Geographical location where Services are to be performed.
Tailings	Fine grained waste product from ore beneficiation or mineral process.
Underdrainage system	Material or combination of materials designed and constructed to allow interception and recovery of permeating liquors within a tailings storage facility.
UV	Ultraviolet light.

Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards or codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All Work performed and materials furnished shall conform to the following.

3.2.1 International Codes and Standards

3.2.1.1 ASTM – American Society for Testing and Materials

ASTM C444M-03	Standard Specification for Perforated Concrete Pipe (Metric)
ASTM D6707	Standard Specification for Circular-Knit Geotextile for Use in Subsurface Drainage Applications
ASTM F758	Standard Specification for Smooth-Wall Poly(Vinyl Chloride) (PVC) Plastic Underdrain Systems for Highway, Airport, and Similar Drainage
ASTM F2648/F2648M	Standard Specification for 2 to 60 inch (50 to 1500 mm) Annular Corrugated Profile Wall Polyethylene (PE) Pipe and Fittings for Land Drainage Applications

3.2.2 Alternative Codes

Alternative codes may be used for supply and installation of underdrainage if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0004	Stormwater Drainage and Gravity Sewers
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. MATERIALS

4.1 General

Supply of underdrainage pipe and fittings shall be carried out in accordance with the requirements of the following sections:

- Corrugated Perforated Drainage Pipe Material
- Packaging and Handling
- Transportation, Loading and Unloading, and
- Quality Assurance.

Unless otherwise specified, all underdrainage pipe shall be supplied complete with fittings and geotextile filter fabric. A prequalification document for materials to be supplied complete with technical details and justification for use shall be submitted to the Employer for review and approval prior to procurement.

4.2 Corrugated Perforated Drainage Pipe Material

4.2.1 Pipe Strength

Underdrainage pipe shall comply with ASTM F758 (PVC pipe) or ASTM F2648/F2648M (PE pipe).

The intended method of installation for the coiled pipe is for the pipe to be ploughed into place typically using a mandrill fitted to a D8 (or similar) dozer. Rigid pipe is intended to be installed within a trench and backfilled. The pipe, filter fabric and the manufacturer's recommended method of coupling shall have sufficient tensile strength to accommodate the applicable method of installation.

4.2.2 Pipe Perforations

Pipe perforations in underdrainage pipe shall be in the form of slots and shall be made in the bottom of the corrugation valleys.

The perforations shall comply with ASTM C444M-03 and shall meet the following requirements:

- Minimum clear water opening of 1500 mm^2/m
- Maximum perforation width of 1.5 mm, and

- Perforation spacing to be uniform over the length of the pipe
- Finished edges to slots shall be clean, flush and free from surplus material resulting from the fabrication process.

4.2.3 Filter Fabric

Underdrainage pipe shall be supplied with polypropylene filter sock conforming to ASTM D6707. Filter fabric shall be manufactured from 150 denier (approx. 17 tex) grade yarn (or approved equivalent), meeting the representative large opening size and opening index specified for the project.

The filter fabric shall perform satisfactorily for the type of liquor/tailing to be stored in the tailing storage area.

A prequalification document for filter fabric suitable for the tailing to be stored shall be submitted to the Employer for review and approval prior to procurement.

4.3 Packaging and Handling

4.3.1 Drain-Pipe Coils

- Polypropylene Filter Fabric to be stencil marked with the word "polypropylene" (text height minimum of 50 mm, preferably blue) at the end of the coil
- Drain-Pipe coils (100 m lengths) shall be secured with flat nylon strapping of minimum width of 10 mm, or equivalent, in order to minimise any damage to the piping and/or filter fabric during handling, transport and storage
- The Contractor shall be responsible for ensuring that the methods used for strapping the rolls, and stacking and securing the rolls within the container shall not cause any permanent denting or deformation to the pipe or any damage to the Filter Fabric
- Drain-Pipe rolls shall be double 'wrapped' with 100 micron (minimum) HDPE (or approved equivalent). Wrapping to be stencil marked with the month of manufacture in a minimum of six locations per roll e.g. "Jan 2010" (text height minimum of 100 mm, preferably white).
- The Employer reserves the right to inspect and accept/reject the Drain-Pipe coils at any point of storage/handling, transport, and implementation process.

4.3.2 Rigid Pipe (with Geotextile)

- The Contractor shall comply with the manufacturers' recommendations for transporting, handling, storing, and protecting products
- The Contractor shall avoid shock or damage to the pipe
- The Contractor shall take measures to prevent damage to the geotextile
- Pipe shall not be stacked higher than recommended by pipe manufacturer
- Gaskets for mechanical and push-on joints shall be stored in cool, dry location out of direct sunlight and not in contact with petroleum products

- The Contractor shall provide concrete cradles and encasement for pipeline where indicated on drawings
- The Employer reserves the right to inspect and accept/reject the pipe.

4.4 Transportation, Loading and Unloading

The Contractor shall be solely responsible for obtaining transport permits and clearance (if required) from road or rail authorities for the transporting and delivery of materials and equipment as well as the compliance with the requirements of such bodies.

The Contractor shall be responsible for off-loading all the material and equipment at the Site as well as the storing of such material in a depot as nominated by the Employer.

Selection of the specific loading arrangement and unloading methods shall be negotiated in advance with receiving Site personnel, taking into account available unloading plant and constraints.

The Contractor shall be responsible for ensuring that the pipe is not damaged within any phases of transportation, loading and unloading, until received and accepted by the Employer in writing.

The Employer reserves the right to inspect and accept/reject the loading and transportation procedures. Final acceptance will not be given until the materials have been unloaded at the Site.

4.5 Quality Assurance

4.5.1 General

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements

4.5.2 Quality Assurance of Materials

The Contractor shall provide documentation demonstrating that the pipe, Filter Fabric and fittings comply with the specified Standards.

The Contractor shall provide documentation demonstrating corporate and manufacturing accreditation to ISO 9001 (or approved equivalent).

The Contractor shall provide samples of the materials proposed, complete with technical data detailing:

- Dimensional tolerance of slots sizes (width, length)
- The total area of perforations (mm^2 per meter length)
- Representative Large Opening Size and Opening Index (in respect of Filter Fabric), and
- Other relevant technical information required by the Employer (e.g. UV tolerance/degradation, specific dates of manufacture).

In the absence of any other test frequency in referenced documents under Sections 3.2.1 and/or 3.2.2 the test frequencies as in Table 4-1 shall apply.

Table 4-1 : Test Frequencies

Tests	Characteristics	Requirement	Frequency
Type Test (TT) – Pipes	Performance	Bending; Straightening; High and low temperature impact resistance; Elongation; Pipe stiffness; Perforation deformation; and Joint separation	At any new formulation or change in design
Batch Release Test (BRT) – Pipes	Freedom from defects Dimensions	(Visual inspection) (Diameter, Length)	Once per production shift
Process Verification Test	Dimensions	Perforation dimensions and area	At change in design but not less than once per year
	Performance	Low temperature impact resistance	Not less than once per year

4.6 Tolerances

- All tolerances shall comply to a minimum of those specified in the Standards referenced in Section 3.2.1
- Drain-Pipe length as specified by the manufacturer shall be within –0 mm and +500 mm of the length specified based on an average of three rolls.

5. INSTALLATION

5.1 General

Piping shall be installed as per the manufacturer's specification and in accordance with the drawings. The method of installation for the coiled pipe is for the pipe to be ploughed into place typically using a mandrill fitted to a D8 (or similar) dozer. Rigid pipe is intended to be installed within a trench and backfilled.

Slotted pipe shall be installed as shown on the drawings, and shall be placed centrally within the trench, with perforation orientation as shown on the drawings (where applicable).

Any required joints in slotted pipe shall be made in accordance with the manufacturer's instructions.

5.2 Backfill Material

Backfill and/or filter aggregate shall be placed over the slotted pipe to the depth shown on the drawings, and smoothed to a level profile. Where specified on the drawings, to avoid post-construction settlement, the filter aggregate material shall be compacted to its full depth to achieve effective mechanical interlock between particles.

5.3 Flushing

After completion of backfilling, underdrains shall be flushed in the presence of the Employer with sufficient clean water until only clean water discharges at the outlet.

5.4 Underdrainage

The Contractor shall provide controls to manage the depth of drainpipe such as to avoid damage to the liner.

If, during underdrainage system installation, accidental tearing of the liner occurs, or damage to the liner is suspected, the area shall be isolated, checked and rectified. Pipe installation shall only recommence when the Employer has inspected the repairs. The Contractor shall rework any damaged area of the liner as a result of the underdrainage system installation.

5.5 Jointing

Jointing of pipes shall be kept to a minimum, and where required, shall be made using suitable couplings and in accordance with the drawings and the manufacturer's requirements.

Jointing of coiled pipe shall be carried out as soon as practical after piping has been unrolled to minimise the possibility of damage to the geofabric sock from UV exposure.

Where drainpipe is to be ploughed into position, jointing shall be done prior to installation within the trench to prevent the possibility of un-jointed drainpipe within the sand layer. Also, where drainpipe is to be ploughed into position, the Contractor shall demonstrate that the joint remains sound after installation.

5.6 Backfilling

Backfilling over pipe joints shall only commence after the Employer has inspected the joints.

5.7 Tolerances

- Pipe shall not deviate from the specified line by more than 100 mm and level by more than 50 mm at any point. The alignment of sequential pieces of rigid pipe shall not deviate by more than the manufacturer's specified limits, or in any case, by an amount that compromises performance of the pipe or joint.
- Slope tolerances shall be as under:

Specified Gradient	Tolerance
Flatter than 1:200	0.1%
1:200 to 1:50	0.2%
Steeper than 1:50	0.5%

6. SUBMITTALS

The Contractor shall transmit the following submittals to the Employer:

- A Quality Control Program
- A prequalification document for materials
- A detailed work plan

- A statement attesting to compliance of the Contractor's Work with the requirements of this specification and other Contract Documents
- Site specific Inspection and Test Plan (ITP) and Test Reports for review and acceptance as described in Section 4.5. The Inspection and Test Plans shall clearly identify participation of all parties concerned including but not limited to, the Contractor and Testing Agency. The ITP shall identify contract documents referenced to for activities to be undertaken, as well as all hold points.